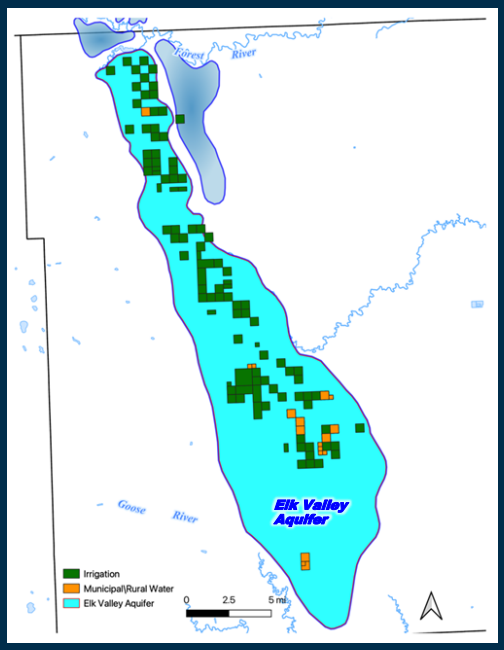
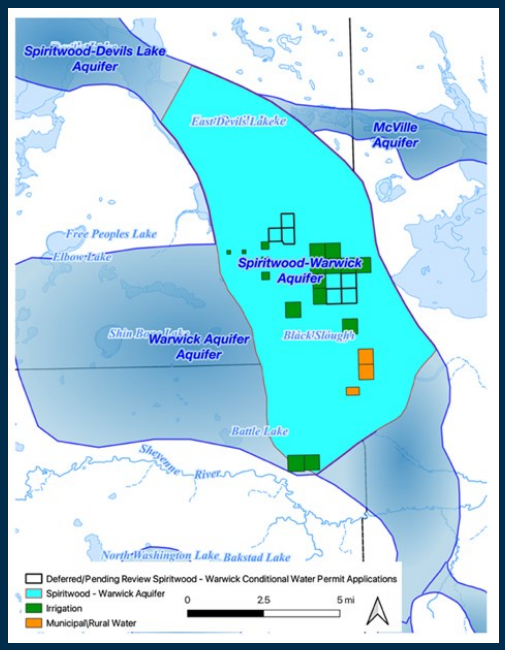
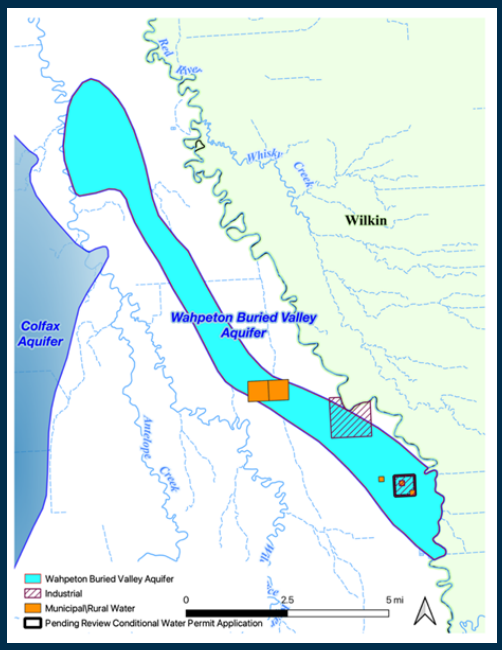
COST PROPOSAL

Transient Groundwater Flow Models for Water Availability and Artificial Recharge Projections

Request for Proposal Number: 770.3000-25-01







Prepared for

NORTH DAKOTA DWR

Prepared by

INTERA

in association with

northwater

OCTOBER 11, 2024

Total Cost

INTERA's total cost for developing transient groundwater flow models for the Elk Valley aquifer, Spiritwood-Warwick aquifer, and Wahpeton Buried Valley aquifer system, and conducting the associated analyses to evaluate aquifer resiliency, MAR, and conditional permits is $547,372.

Itemized Costs

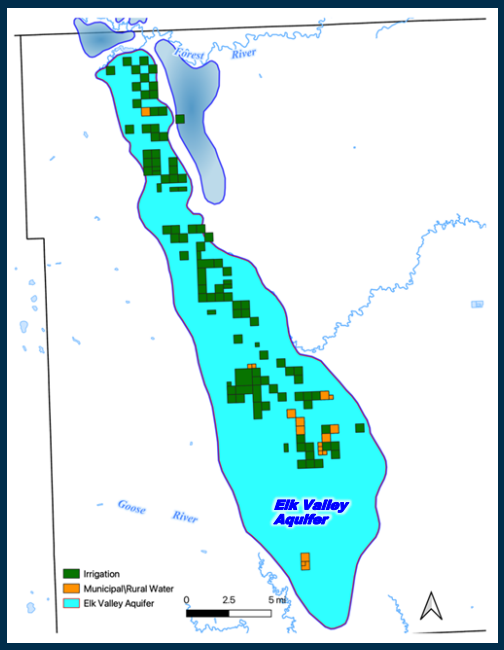
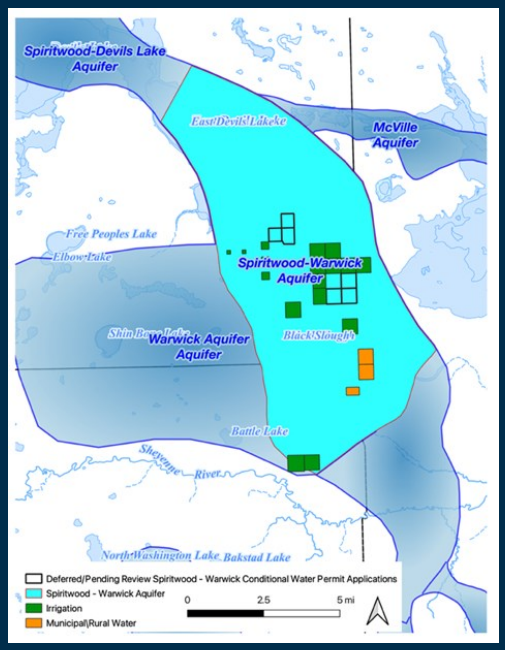
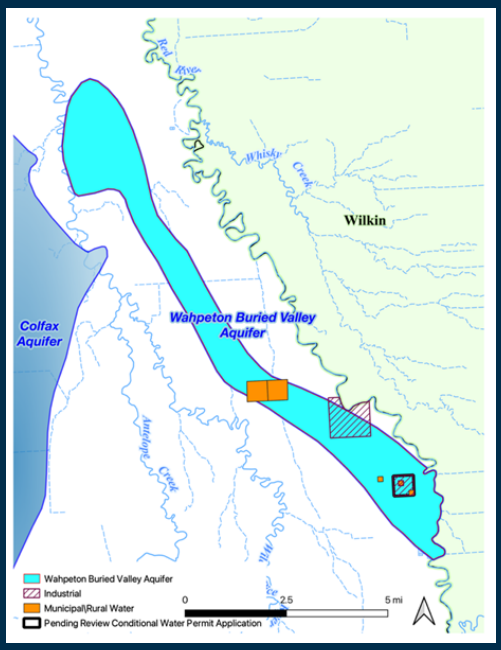
INTERA's detailed itemization of the costs required to complete this project for DWR is provided in the table below.

		Project Phases													
		Planning & Coordination		Numerical Modeling Approach		Numerical Model Development		Modeling Scenarios		Draft Report		Final Report		Total	
		hrs	$	hrs	$	hrs	$	hrs	$	hrs	$	hrs	$	hrs	$
INTERA Labor Categories and Rates															
Principal Eng/Sci I	$306/hr	8	2,448	12	3,672	12	3,672	12	3,672	8	2,448	8	2,448	60	18,360
Principal Eng/Sci II	$274/hr	8	2,492	16	4,384	16	4,384	16	4,384	20	5,480	40	10,960	116	31,784
Principal Eng/Sci III	$257/hr														
Senior Eng/Sci I	$240/hr														
Senior Eng/Sci II	$218/hr														
Senior Eng/Sci III	$206/hr														
Senior Eng/Sci IV	$195/hr	180	35,100	40	7,800	450	87,750	120	23,400	160	31,200	40	7,800	990	193,050
Eng/Sci I	$184/hr	60	11,040	40	7,360	480	88,320	120	22,080	160	29,440	40	7,360	900	165,600
Eng/Sci II	$173/hr														
Eng/Sci III	$150/hr	120	18,000	20	3,000	400	60,000	80	12,000	80	12,000	16	2,400	716	107,400
Eng/Sci IV	$139/hr														
Intern	$93/hr														
Senior Technician	$153/hr														
Technician	$93/hr														
Sr Technical Editor	$153/hr														
Technical Editor	$104/hr														
Sr CAD/Graphics	$132/hr														
CAD/Graphics	$104/hr														
Project Associate	$96/hr														
INTERA Labor Total		376	68,780	128	26,216	1,358	244,136	348	65,536	428	80,568	144	30,968	2,782	**516,194**
Other Direct Costs															
Northwater (subcontractor)			14,450		1,665				1,665		3,330		2,000		23,110
subcontractor markup (15%)			2,168		250				250		500		300		6,222
Travel			2,000								2,000				4,000
travel expense markup (15%)			300								300				600
ODC Total			40,032		1,915				1,915		6,130		2,300		**31,178**
														PROJECT TOTAL	**547,372**

Value Added Options:
1. Soil water balance - $19,500
2. Data-worth analysis - $31,200
3. Additional uncertainty analysis - $32,960

Value-Added Options

As discussed in our technical proposal, and shown at the bottom of the table above, INTERA has identified three value-added options—a soil water balance, a data-worth analysis, and additional uncertainty analysis. The cost of these options are included on the table.

INTERA Incorporated

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